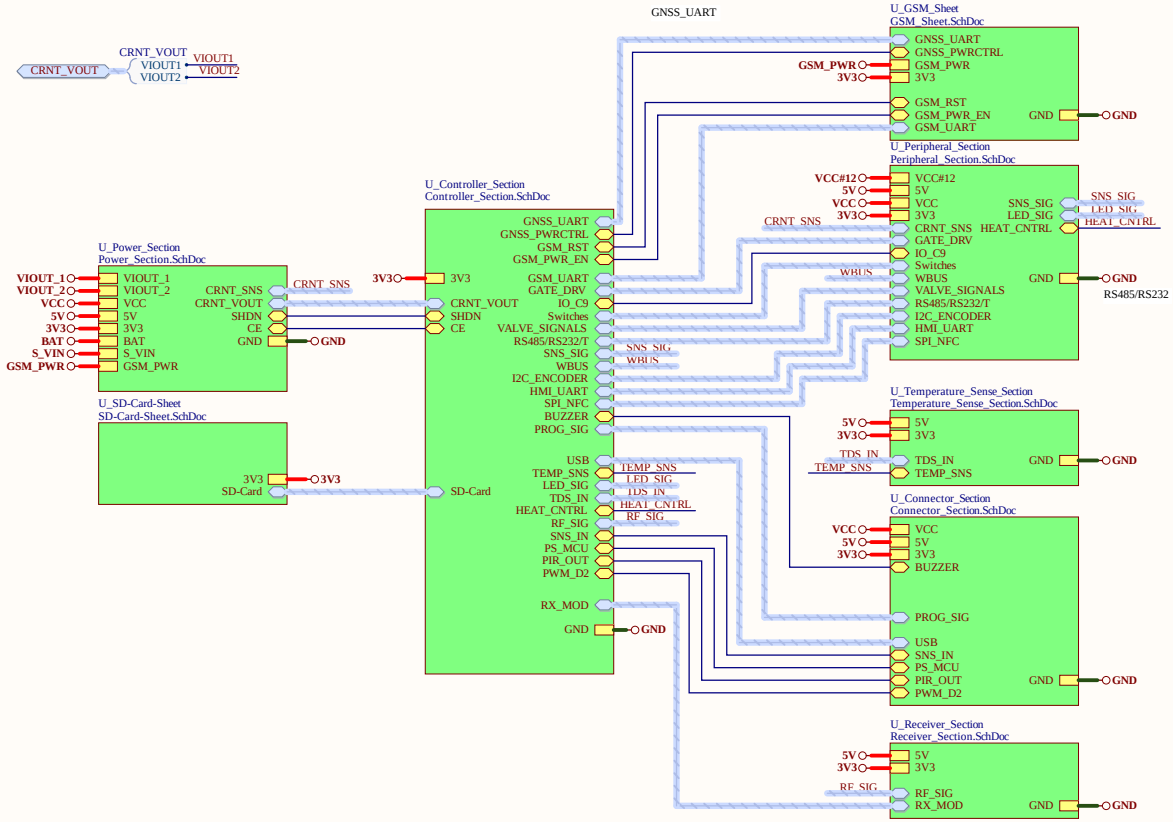
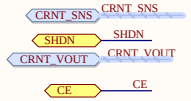
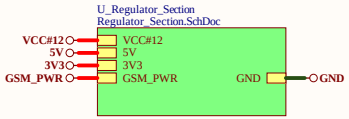
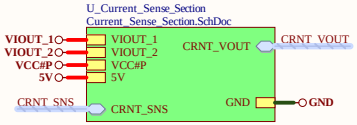


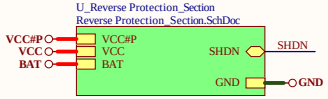
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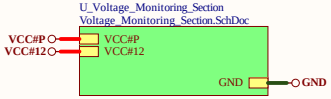
Title: Top_Sheet.SchDoc	
File: GTV3.3_Board.PrjPcb	
Date: 2/10/2024	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 17

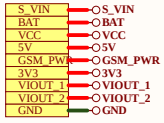


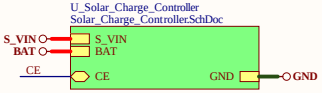










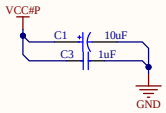


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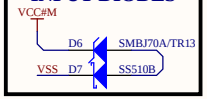
Title: Power_Section.SchDoc	
File: GTV3.3_Board.PrjPcb	
Date: 2/10/2024	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 17

SHDN SHDN

DECOUPLING CAPACITOR

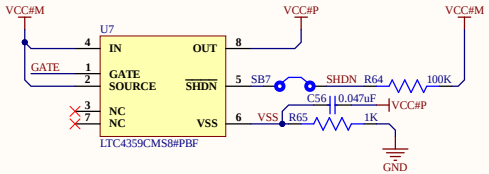


INPUT DIODES

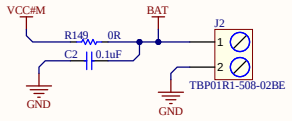


BAT — **BAT**
VCC — **VCC**
VCC#P — **VCC#P**
GND — **GND**

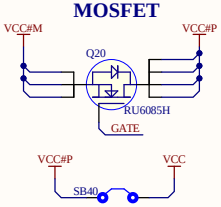
REVERSE POLARITY PROTECTION IC



BATTERY / MAIN INPUT



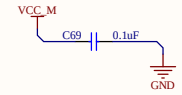
MOSFET



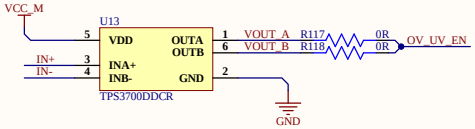
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Title: Reverse Protection_Section.SchDoc	
File: GTV3.3_Board.PrjPcb	
Date: 2/10/2024	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 17

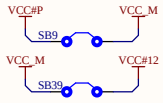
DECOUPLING CAPACITOR



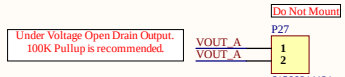
UNDER AND OVER VOLTAGE PROTECTION IC



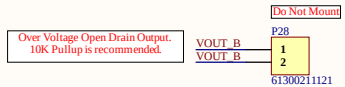
VOLTAGE TO MONITOR



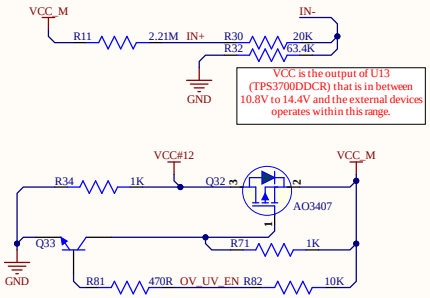
UNDER VOLTAGE OUTPUT

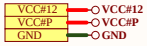


OVER VOLTAGE OUTPUT



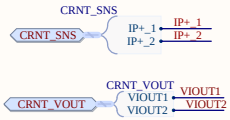
SUPPORTING CIRCUITRY

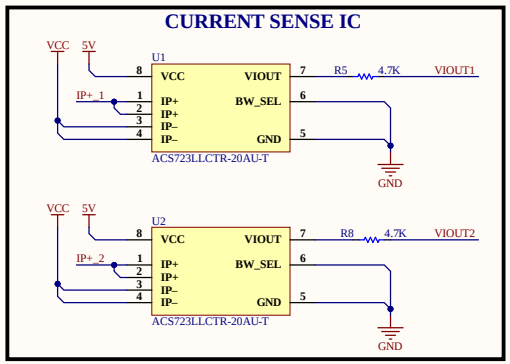




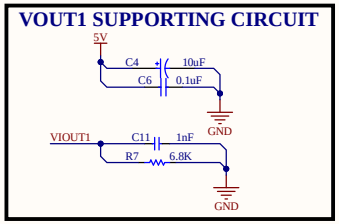
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Template\espark-consulting-group-800x300.jpg.
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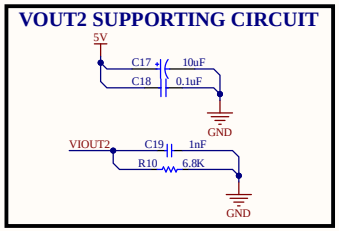
Title: Voltage_Monitoring_Section.SchDoc	
File: GTV3.3_Board.PrjPcb	
Date: 2/10/2024	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 17

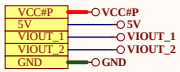




Two current sensing IC's are used to provide support to monitor current of two Pumps. Whereas currently we have only one pump in our V3 Controller.



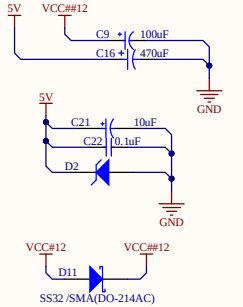




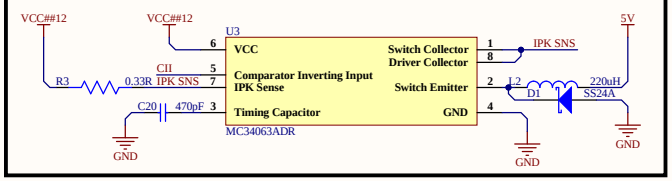
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Title: Current_Sense_Section.SchDoc	
File: GTV3.3_Board.PrjPcb	
Date: 2/10/2024	Page Num: 5
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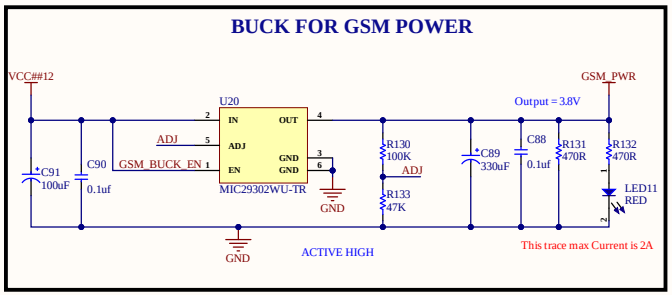
DECOUPLING CAPACITOR



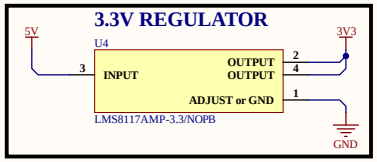
5V BUCK REGULATOR



BUCK FOR GSM POWER

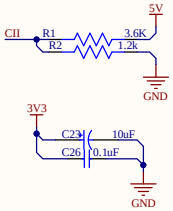


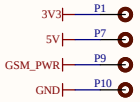
3.3V REGULATOR



VOUT = 1.25 (1+ R2/R1)
R2=3.6K
R1=1.2K

SUPPORTING CIRCUIT

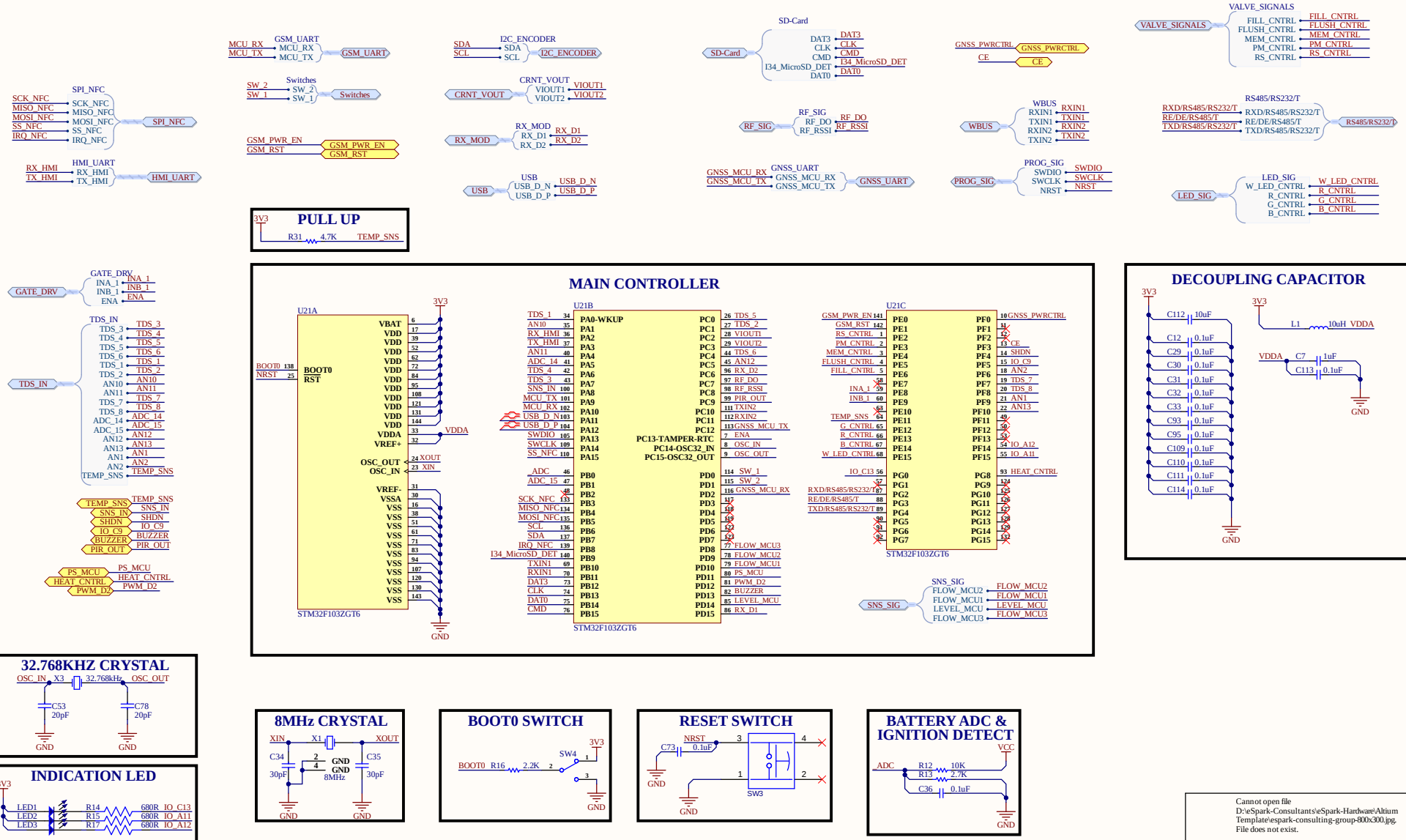






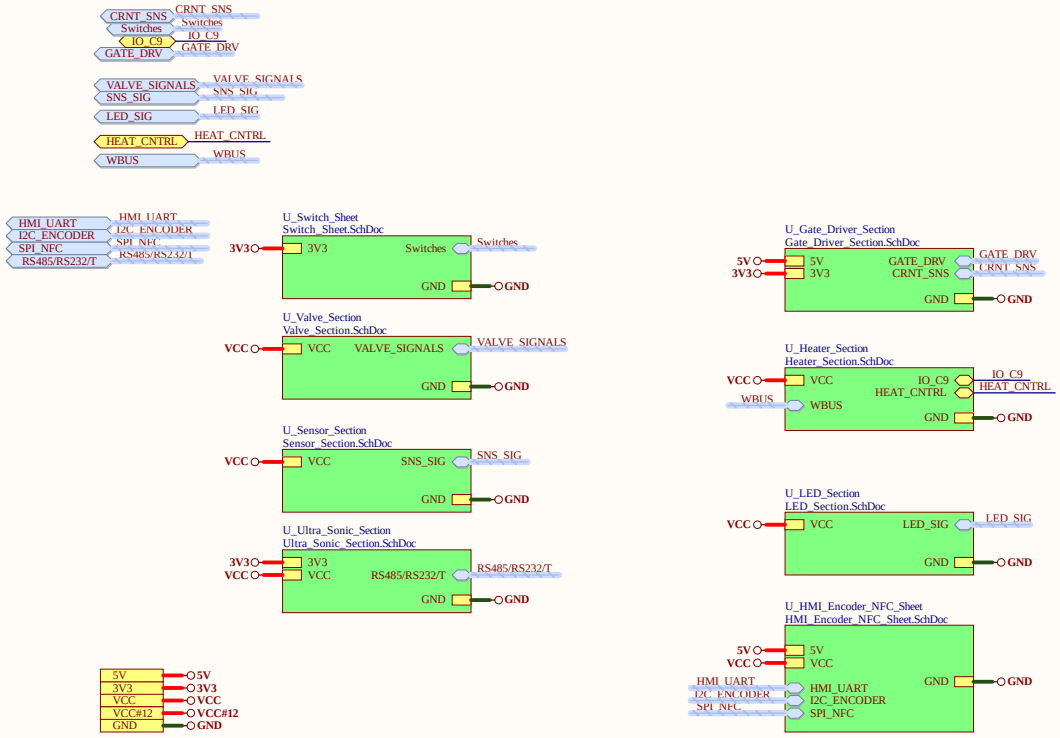
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Title: Regulator_Section.SchDoc	
File: GTV3.3_Board.PrjPcb	
Date: 2/10/2024	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 17



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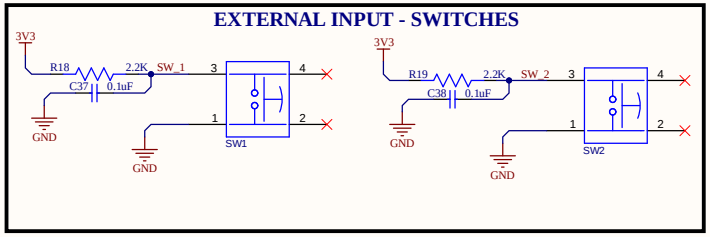
Title: Controller_Section.SchDoc	
File: GTV3.3_Board.PrjPcb	
Date: 2/10/2024	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 17



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Title: Peripheral_Section.SchDoc	
File: GTV3.3_Board.PrjPcb	
Date: 2/10/2024	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 17

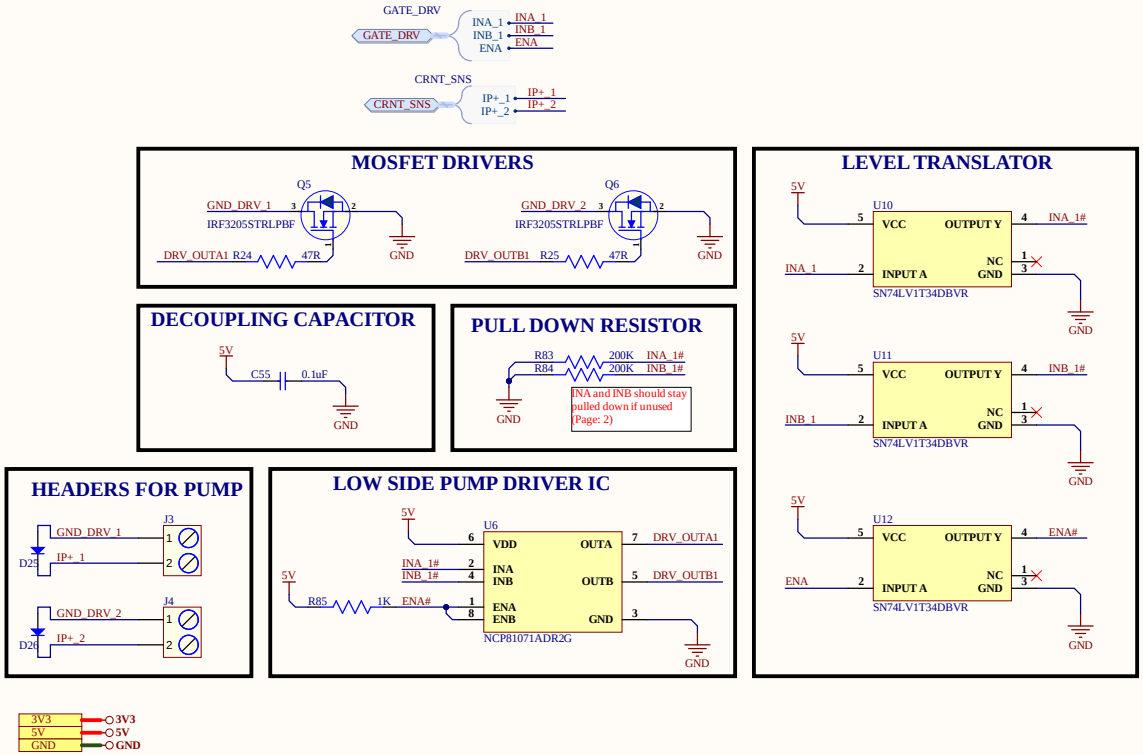






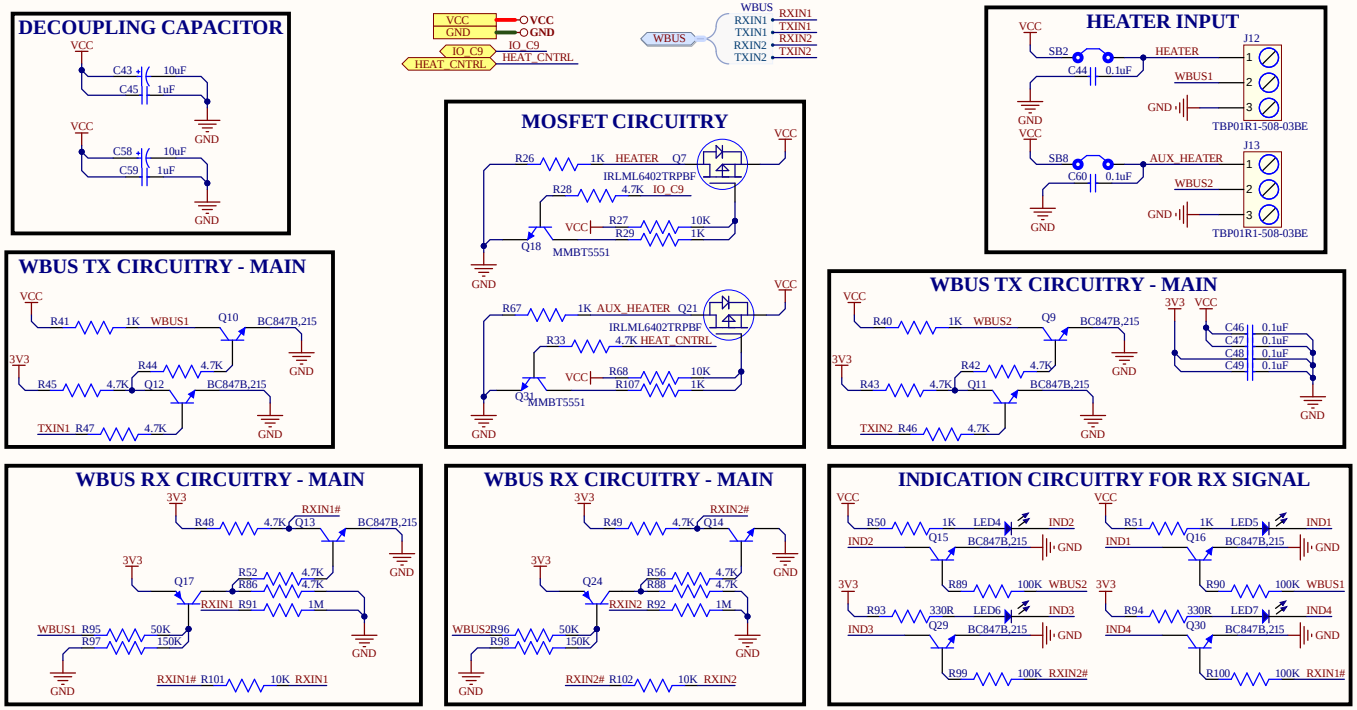
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Title: Switch_Sheet.SchDoc	
File: GTV3.3_Board.PrjPcb	
Date: 2/10/2024	Page Num: 9
Drawn By: Huzaifa Ali	Page Total: 17



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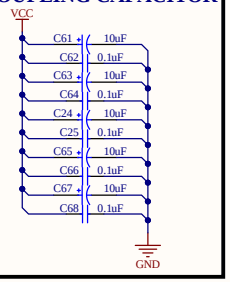
Title: Gate_Driver_Section.SchDoc	
File: GTV3.3_Board.PrjPcb	
Date: 2/10/2024	Page Num: 10
Drawn By: Huzaifa Ali	Page Total: 17



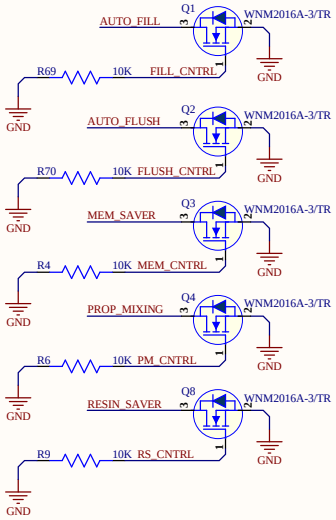
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Title: Heater_Section.SchDoc	
File: GTV3.3_Board.PrjPcb	
Date: 2/10/2024	Page Num: 11
Drawn By: Huzaifa Ali	Page Total: 17

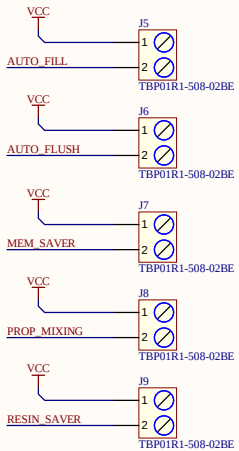
DECOUPLING CAPACITOR



SOLENOID CONTROL CIRCUIT



SOLENOID VALVE INPUTS



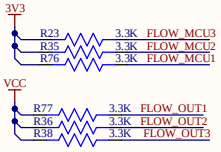




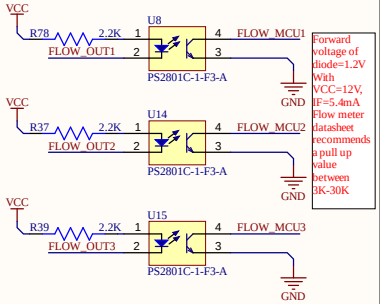
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Title: Valve_Section.SchDoc	
File: GTV3.3_Board.PrjPcb	
Date: 2/10/2024	Page Num: 12
Drawn By: Huzaifa Ali	Page Total: 17

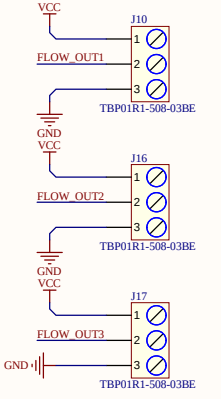
PULL UP RESISTORS



OPTO ISOLATION FOR FLOW METER



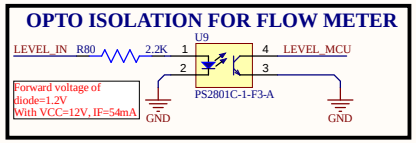
FLOW METER INPUT



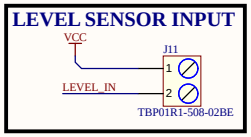
SUPPORTING CIRCUIT



OPTO ISOLATION FOR FLOW METER



LEVEL SENSOR INPUT

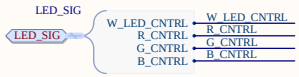


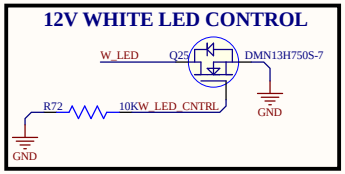


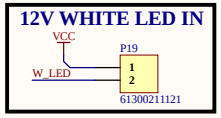


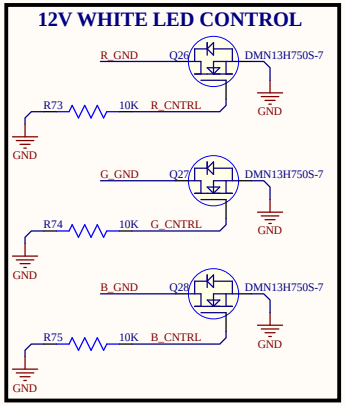
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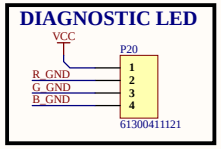
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Date: 2/10/2024	Page Num: 13
Drawn By: Huzaifa Ali	Page Total: 17







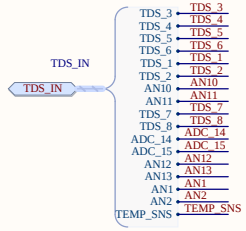




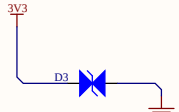


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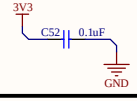
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Date: 2/10/2024	Page Num: 14
Drawn By: Huzaifa Ali	Page Total: 17



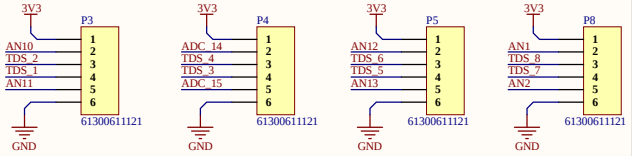
PULL UP RESISTORS



DECOUPLING CAP

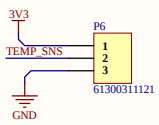


4 X TDS INPUTS



These TDS headers connect with TDS header on V3 Controller PCB via Female to Female Cable.

TEMP SENSOR DS18B20

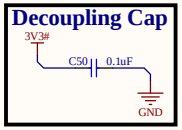




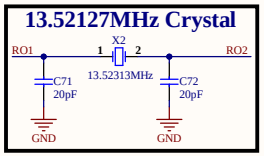
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Date: 2/10/2024	Page Num: 15
Drawn By: Huzaifa Ali	Page Total: 17

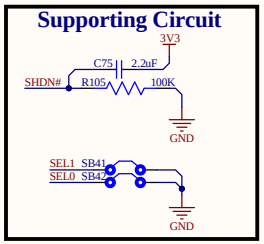
Decoupling Cap

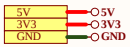


13.52127MHz Crystal



Supporting Circuit

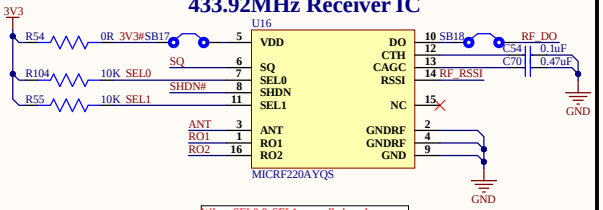






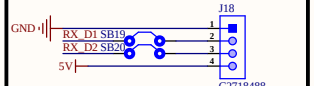


433.92MHz Receiver IC

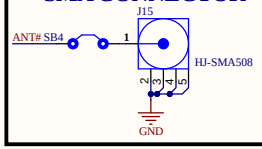


When SEL0 & SEL1 are pulled up, low pass filter BW = 13000Hz
Max encoded bit rate =20Kbps
Pulse width @433.92MHz=38us
CAGC Valu= 0.47uF
FREF= FRF/(32+87/1000)
FREF= 13.52323MHz (@433.92MHz FRF)

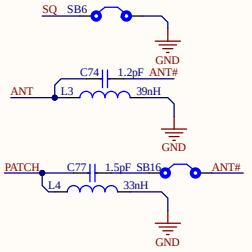
RX 433.93MHz MODULE



SMA CONNECTOR

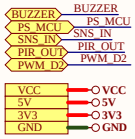


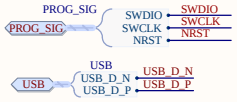
Z-MATCHING CIRCUITRY

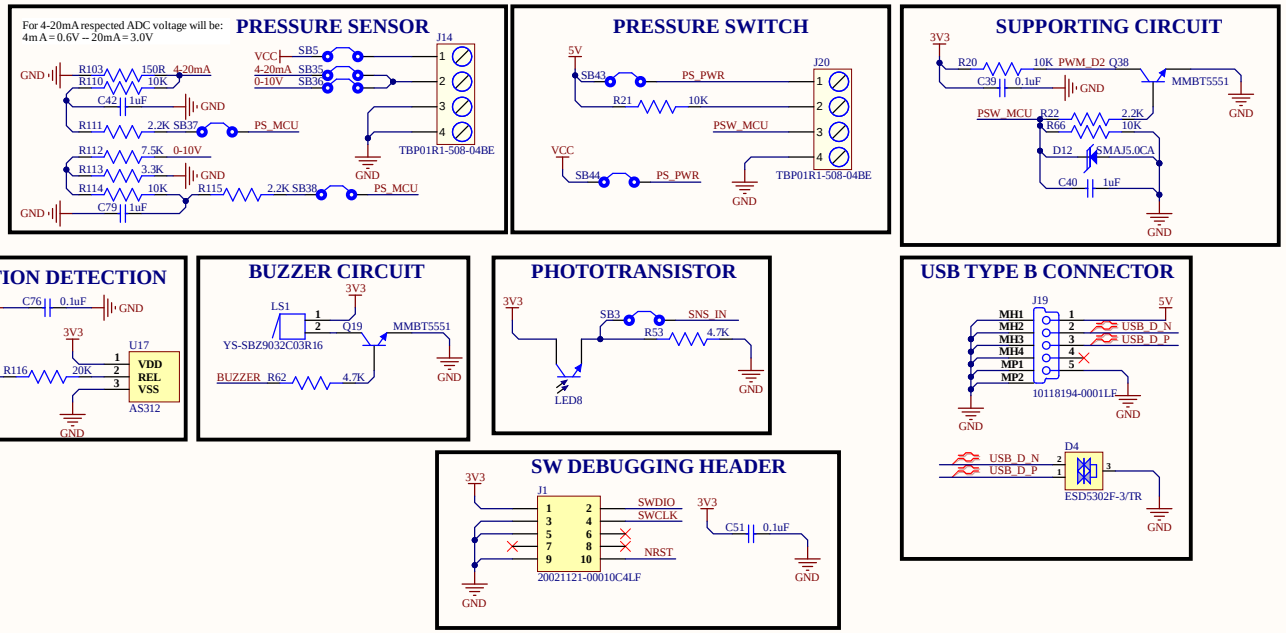


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Drawn By: Huzaifa Ali	Page Total: 17



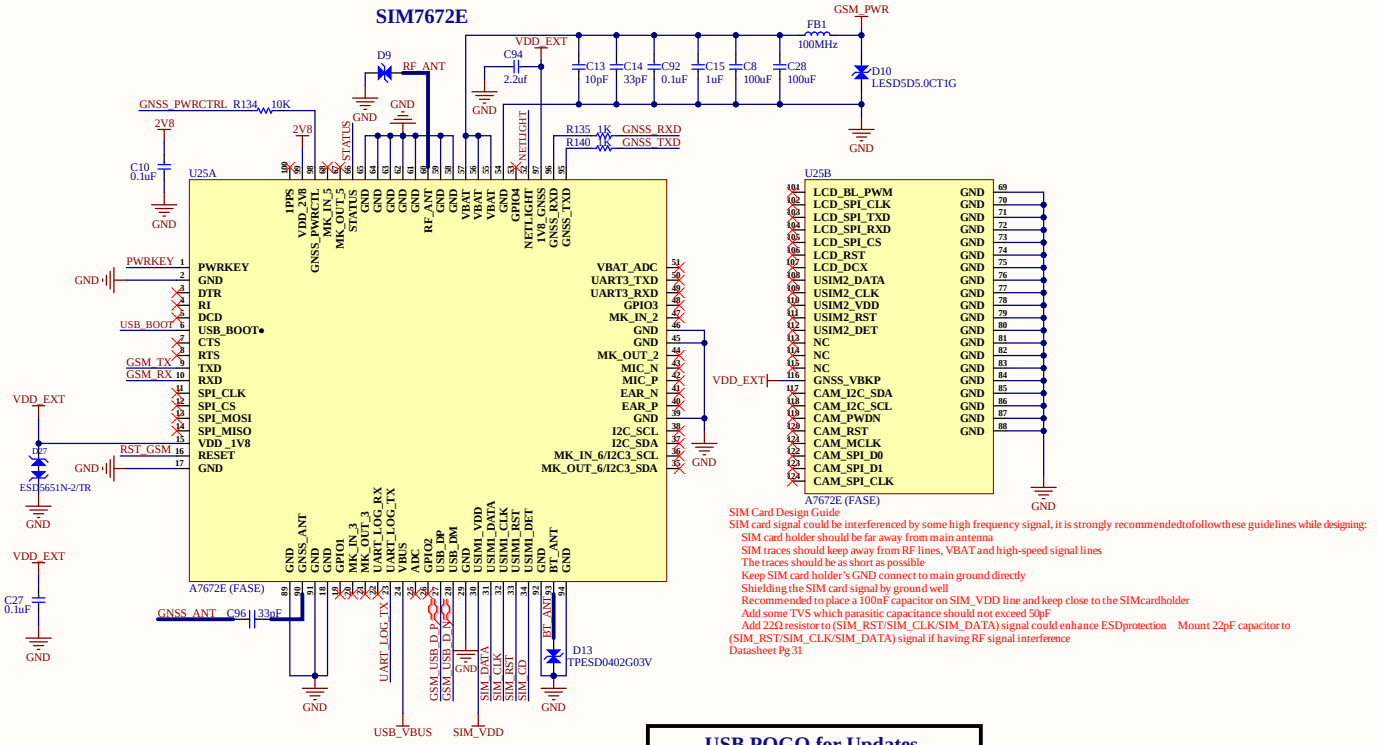




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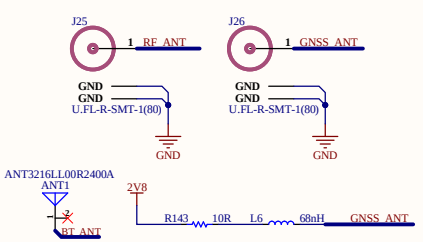
SIM7672E



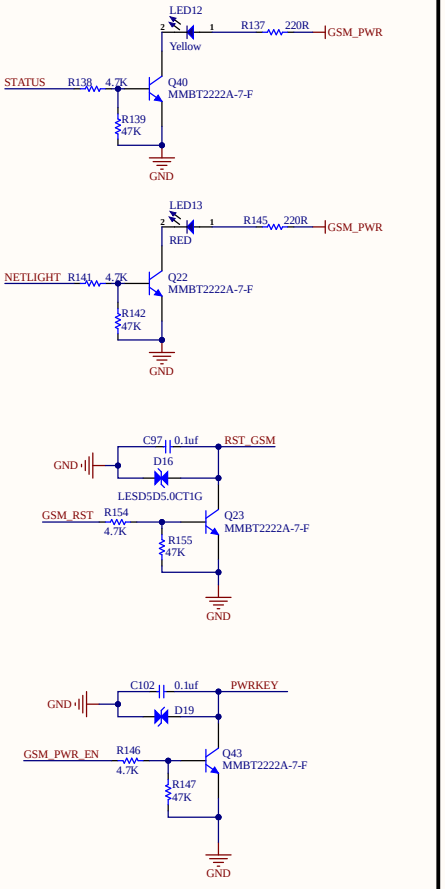
SIM Card Design Guide
SIM card signal could be interferenced by some high frequency signal, it is strongly recommendedtofollowthese guidelines while designing:
- SIM card holder should be far away from main antenna
- SIM traces should keep away from RF lines, VBAT and high-speed signal lines
- The traces should be as short as possible
- Keep SIM card holder's GND connect to main ground directly
- Shielding the SIM card signal by ground well
- Recommended to place a 100nF capacitor on SIM_VDD line and keep close to the SIMcardholder
- Add some TVS which parasitic capacitance should not exceed 50pF
- Add 22Ω resistor to (SIM_RST/SIM_CLK/SIM_DATA) signal could enhance ESDprotection Mount 22pF capacitor to (SIM_RST/SIM_CLK/SIM_DATA) signal if having RF signal interference

Datasheet Pg 31

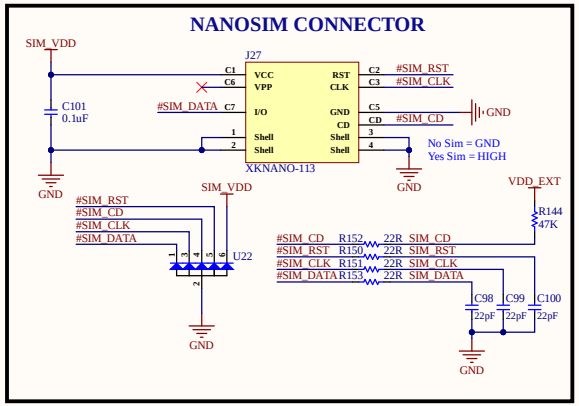
ANTENNA



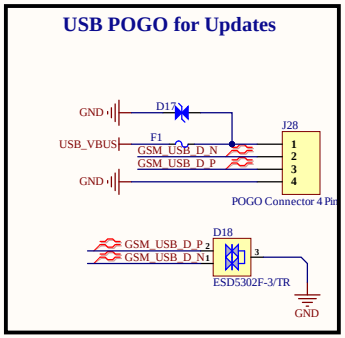
GSM_COMMANDS



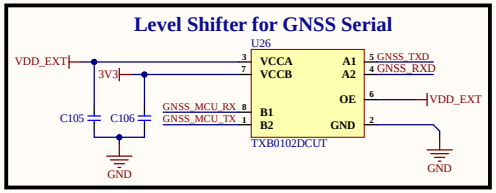
NANOSIM CONNECTOR



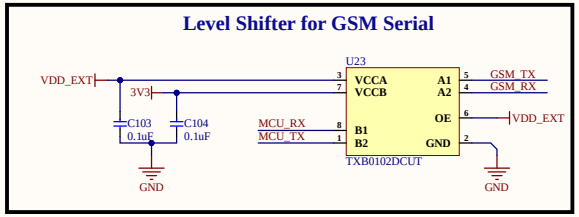
USB POGO for Updates

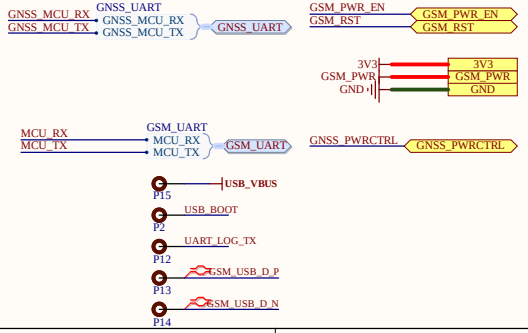


Level Shifter for GNSS Serial



Level Shifter for GSM Serial

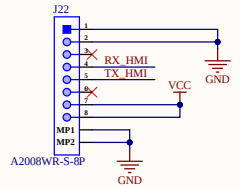




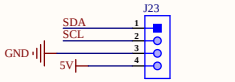
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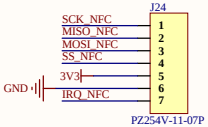
HMI INPUT STONE

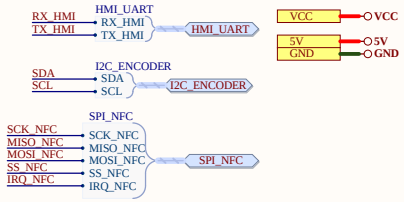


ENCODER INPUT



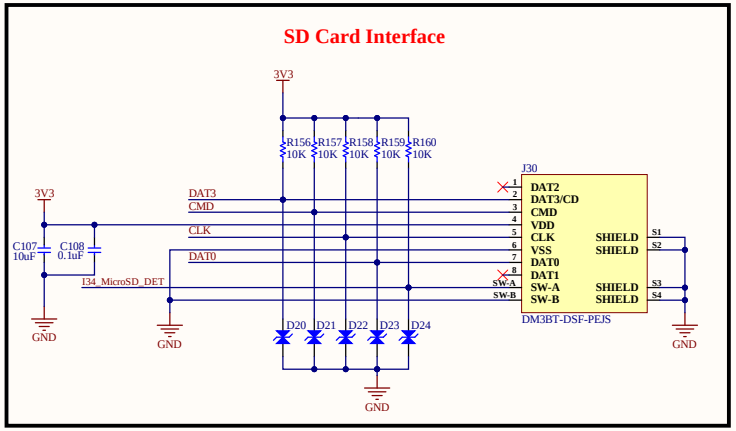
NFC INPUT

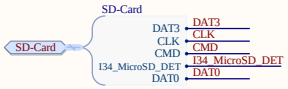




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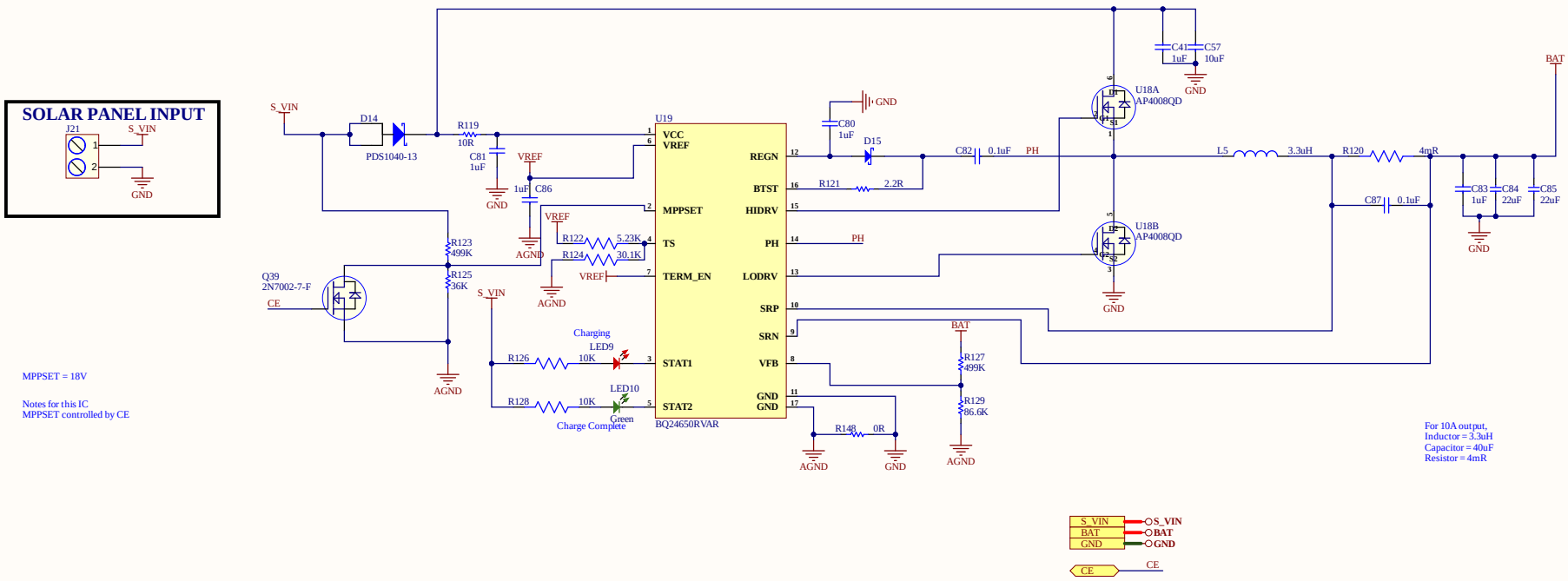






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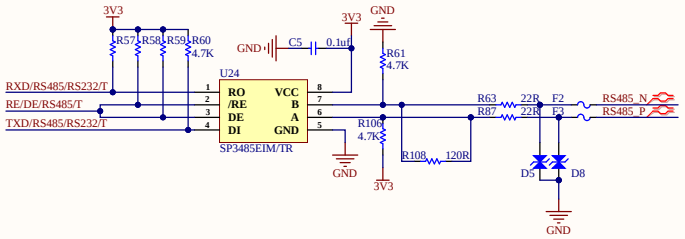
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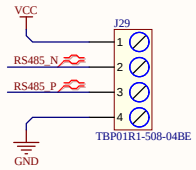
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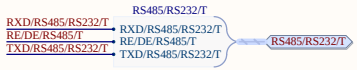
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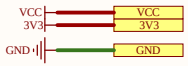
RS485 to UART



ULTRA-SONIC SENSOR







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